HINT & SOLUTION OF ELASTICITY & VISCOSITY
EXERCISE-1
PART - I

A-1. (a) $\sigma = \frac{F}{A} = \frac{3\times10^4}{3.6\times10^{-4}} = \frac{30}{36}\times10^8 = \frac{5}{6}\times10^8 < 7.7\times10^8$ N/m^2

hence it will not break.

(b) $\Delta\ell = \frac{FL}{AY} = \frac{3\times10^4\times20\times10^{-2}}{3.6\times10^{-4}\times1.5\times10^{10}} = \frac{40\times10^{-4}}{3.6} = \frac{10}{9}\times10^{-3}$ m

A-2. $\Delta\ell_S = \frac{F\ell}{Y_S A}$ $\Delta\ell_C = \frac{F\ell}{Y_C A}$

$\Delta\ell_S - \Delta\ell_C = 0.5$

$\frac{F\ell}{A}\left(\frac{1}{Y_s}-\frac{1}{Y_c}\right) = 0.5$

$\frac{F\ell}{A} = \frac{Y_c Y_s \times 0.5}{(Y_c - Y_s)}$

$\frac{F\ell}{A} = 15\times10^{11}$

$\Delta\ell_S = \left(\frac{F\ell}{A}\right)\frac{1}{Y_s} = \frac{15\times10^{11}}{2\times10^{12}} = 0.75$ cm

$\Delta\ell_C = \left(\frac{F\ell}{A}\right)\times\frac{1}{Y_c} = \frac{15\times10^{11}}{12\times10^{11}} = 1.25$ cm

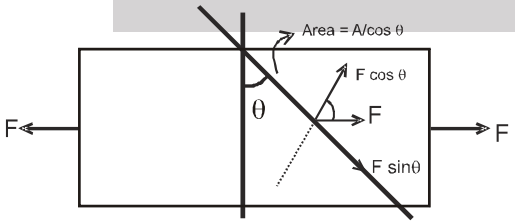
A-3. $a = \frac{40}{12} = \frac{10}{3}$ m/s^2

$T_1 = \frac{40}{3}$ N $\qquad$ $T_2 = \frac{80}{3}$ N

strain in wire 1 $= \frac{40}{3\times5\times10^{-7}\times2\times10^{11}} = \frac{4}{3}\times10^{-4}$

strain in wire 2 $= \frac{80}{3\times5\times10^{-7}\times2\times10^{11}} = \frac{8}{3}\times10^{-4}$

B-1.



(a) tensile stress $= \frac{F\cos\theta}{A/\cos\theta} = \frac{F\cos^2\theta}{A}$

(b) shearing stress $= \frac{F\sin\theta}{A/\cos\theta} = \frac{F}{A}\sin\theta\cos\theta$

(c) for max. tensile stress

$\theta = 0^o$

(d) for max. shearing stress

$\theta = 45^o$

C-1. $B = \dfrac{P}{\dfrac{\Delta V}{V}} = \dfrac{100\text{atm}}{10 \times 10^{-6}} = 10^7\ \text{atm}$

D-1. $K = \dfrac{AY}{\ell} = \dfrac{1\times10^{-4}\times1\times10^{11}}{0.2} = 5 \times 10^7$

$U = \dfrac{F^2}{2K} = \dfrac{(5\times9.8)^2}{2\times5\times10^7} = \dfrac{49\times49}{10^8} = 2.40 \times 10^{-5}\ \text{J}$

D-2. $W = \dfrac{1}{2}(F_2 x_2 - F_1 x_1)$

$= \dfrac{1}{2}(4 \times 9.8 \times 10^{-3} - 2 \times 9.8 \times 0.6 \times 10^{-3})$

$= 2 \times 9.8 \times 10^{-3} \times 0.7$

$= 13.72 \times 10^{-3}\ \text{J}$

E-1. $v = \dfrac{2}{9\eta} r^2 . (\rho_0 - \rho_w) g$

$= 180$ m/sec.

$h = \dfrac{32400}{2\times9.8} = \dfrac{81}{49}\times10^3\ \text{m}$

PART - II

A-1. $d = 4\text{mm}$

$Y = 9 \times 10^{10}\ \text{N/m}^2$

$\dfrac{F}{A} = Y\dfrac{\Delta\ell}{\ell}$

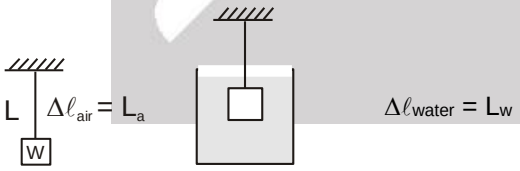
$F = AY\dfrac{\Delta\ell}{\ell}$

$= \pi(2x10^{-3})^2 \text{ x } 9 \times 10^9 \text{ x } \dfrac{1}{100}$

$= \pi \text{ x } 4 \text{ x } 10^{-6} \text{ x } 9 \times 10^7$

$= 360\ \pi\ \text{N}$

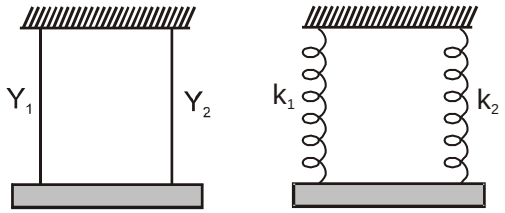
A-2.



$L_a = \dfrac{WL}{YA}$ $\quad L_w = \dfrac{\left[W - \dfrac{W}{\rho_o}\rho_w\right] L}{YA} = \dfrac{W[1 - \dfrac{\rho_w}{\rho_o}] L}{YA}$

$\dfrac{L_a}{L_w} = \left[1 - \dfrac{\rho_w}{\rho_o}\right] \Rightarrow \dfrac{\rho_o}{\rho_w} = \dfrac{L_a}{L_a - L_w}$

A-3.

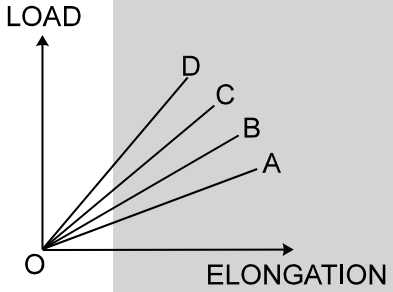


$K_{eq} = K_1 + K_2$

$$\frac{Y2A}{\ell} = \frac{Y_1A}{\ell} + \frac{Y_2A}{\ell}$$

$$Y = \frac{Y_1 + Y_2}{2}$$

A-4.



$$\frac{F/A}{\Delta\ell/\ell} = Y$$

$$\frac{F}{\Delta\ell} = \frac{Y\pi r^2}{\ell}$$

$$\Rightarrow \quad \frac{F\ell}{Y\pi} \times \frac{1}{\Delta\ell} = r^2$$

$\Rightarrow$ Y & ℓ are same for all then

For same load $r \propto \frac{1}{\sqrt{\Delta\ell}}$

B-1. $F = \eta A \frac{x}{h} = 0.4 \times 10^{11} \times 1 \times .005 \times \frac{.02 \times 10^{-2}}{1} = 4 \times 10^4$ N

C-1. $\frac{\Delta V}{V} = \frac{\Delta P}{B} = \frac{1 \times 10^5}{1.25 \times 10^{11}} = 8 \times 10^{-7}$

D-1. $V = 1/2\ K(2)^2$
$V' = 1/2\ K(10)^2$
then $V' = 25V$

D-2. $K = \frac{AY}{\ell}$, $K' = \frac{4AY}{\ell/2} = 8K$

$$\frac{U}{2} = \frac{\frac{1}{2} \times 8K \times \Delta\ell^2}{\frac{1}{2} \times K \times \Delta\ell^2} \Rightarrow \quad U = 16\ J$$

E-1. (i) $v = 5 \times 10^{-4}$ m/s

$$v = \frac{2}{9\eta} r^2 \rho g$$

$$r^2 = \frac{5 \times 9 \times 18 \times 10^{-5} \times 10^{-4}}{2 \times 900 \times 10} = 9 \times 10^{-12}$$

$r = 3 \times 10^{-6}$ m

(ii) $v \propto r^2$

$$\frac{v_1}{v} = \frac{r_1^2}{r^2} = \frac{1}{4},$$

$$v_1 = \frac{5 \times 10^{-4}}{4} = 1.25 \times 10^{-4} \text{ m/sec}$$

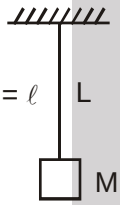
E-2. $v = \frac{2}{9\eta} r^2 . (\rho_0 - \rho_w) g$

E-3. Velocity increases till $F_B + F_V = mg$
F_B = Bouyancy force
F_V = Viscous force

E-4. There will not be any viscous force so velocity will keep on increasing.

PART - III

1.



loss in PE = $Mg\ell$

Elastic PE = $\frac{1}{2} Kx^2$

$= \frac{1}{2} \frac{Mg}{A} \times \frac{\ell}{L} \times AL$

$= MgL/2$

Heat = $MgL - Mg\ L/2$

$= Mg\ L/2$

EXERCISE-2

PART – I

1. $\frac{F}{A} = Y \frac{\Delta\ell}{\ell}$ If Y & $\frac{\Delta\ell}{\ell}$ are constant

$F = AY \frac{\Delta\ell}{\ell}$ $\Rightarrow$ $F \propto A \Rightarrow$ $F' = 4F$

2. $\frac{p_1}{p_2} = \frac{m_1 v_1}{m_2 v_2}$, $m \propto r^3$, $v \propto r^2 \Rightarrow p \propto r^5$ then $\frac{p_1}{p_2} = \frac{1}{32}$

3. $\frac{F}{A} = \eta\frac{x}{h}$

$$\frac{500}{4\times16\times10^{-4}} = 2\times10^{6}\,\frac{x}{4\times\ 10^{-2}} \Rightarrow x = \frac{5\times10^{-2}}{32}\text{m} = 0.156\text{ cm}$$

4. $\ell_B = 2m$ $\quad$ $\ell_S = L$

$A_B = 2\text{ cm}^2$ $\quad$ $A_S = 1\text{ cm}^2$

$\Delta\ell_B = \Delta\ell_S$

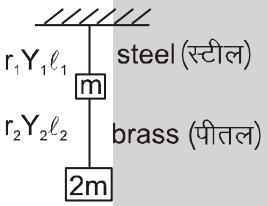
$$\frac{F}{A_B}\frac{\ell_B}{Y_B} = \frac{F}{A_S}\frac{\ell_S}{Y_S}$$

$$L = \frac{A_S Y_S}{A_B Y_B}\ell_B = \frac{1}{2}\text{ x }\frac{2x10^{11}}{1x10^{11}}\text{ x } 2 = 2$$

6. $46.4 \text{ x } 10^{-6}\text{ atm} = \frac{1}{B}$

$$B = \frac{1}{46.4\ \text{ x }\ 10^{-6}} \Rightarrow B = \frac{P}{\Delta v/v} \Rightarrow \frac{\Delta v}{v} = \frac{\Delta p}{B} = 46.4 \text{ x } 10^{-6}$$

7.



$\frac{r_1}{r_2} = b$

$\frac{\ell_1}{\ell_2} = a$

$\frac{Y_1}{Y_2} = c$

$$\Delta\ell_1 = \frac{(3\ \ mg)\ \ \ell_1}{A_1 Y_1}$$

$$\Delta\ell_2 = \frac{(2mg)\ \ \ell_2}{A_2 Y_2}$$

$$\frac{\Delta\ell_1}{\Delta\ell_2} = \frac{3\ell_1}{2\ell_2\ \ A_1 Y_1} \times\ A_2 Y_2 = \frac{3}{2}\ \frac{a}{b^2 c} = \frac{3a}{2b^2 c}$$

8. depth = 200 m

$$\frac{\Delta V}{V} = \frac{0.1}{100} = 10^{-3}$$

density $= 1 \text{ x } 10^3$

$g = 10$

$$B = \frac{\Delta p}{\Delta v/v} = \frac{hg\rho}{\Delta v/v} \qquad \Rightarrow \qquad B = 200 \text{ x } 10 \text{ x } 10^3 \text{ x } 1000 = 2 \text{ x } 10^9$$

9. $\frac{r_1}{r_2} = \frac{1}{2}$

PE (per unit volume) $= \frac{1}{2Y}\left(\frac{F}{A}\right)^2$

$PE \propto 1/A^2$

$\frac{PE_1}{PE_2} = \frac{A_2^2}{A_1^2} = 16 : 1$

10. Magnitude of viscous force is equal to weight and in later case it will act downward.

PART - II

1. $\frac{\left(\frac{\sigma}{2}\right)^2}{2Y} 10x + \frac{\sigma^2}{2Y}(1 - x)\,5 = \frac{40}{100}\left[\frac{\sigma^2}{2Y}10 \times 1\right]$

$2.5x + 5 - 5x = 4$

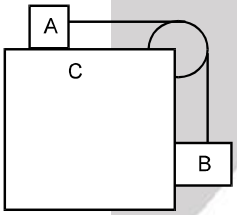
$2.5x = 1$

$x = 40\text{cm}$

2. $\delta = \frac{Fdx}{AY}$, $\Delta\ell = \int \frac{Fdx}{AY}$

$= \int_0^{10} \frac{20 \times 10^3\, dx}{\left(1 + \frac{x^2}{100}\right) \times 2 \times 10^7} = 0.008$ cm

3.



$kx = \mu M_A g$ ..... (i)

$kx = M_B g$ ..... (ii)

$\Rightarrow \quad x = \frac{M_B g}{K} = \frac{1}{100}$

From eq. (i) & (ii)

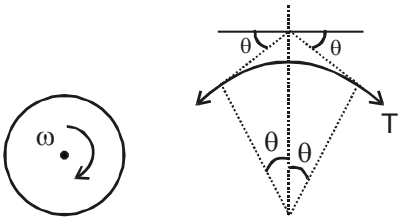
$0.2\, M_A = M_B$

$M_A = 10$ kg

$E = \frac{1}{2}Kx^2 = \frac{1}{2} \times 2000 \times \left(\frac{1}{100}\right)^2$

$E = 10 \times 10^{-2}$

4.



$$2T\sin\theta = \left(\frac{m}{2\pi R}R.2\theta\right)\omega^2R'$$

For small 'θ' -

$$T = \frac{m\omega^2R'}{2\pi}$$

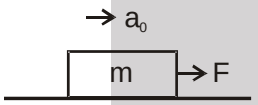
$$\frac{T}{A} = \left(\frac{2\pi R' - 2\pi R}{2\pi R}\right)Y$$

$$\frac{m\omega^2R'}{2\pi A} = \frac{\Delta R}{R}Y, \text{ If } R' \simeq R$$

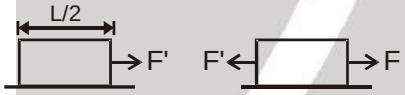
$$\Delta R = \frac{m\omega^2R^2}{2\pi AY} \text{ but } m = A\,2\pi R\,\rho$$

$$\Delta R = \frac{A2\pi R\rho\omega^2R^2}{2\pi AY} = \frac{\rho\omega^2R^3}{Y}$$

5.

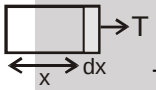


(a)



$$F' = \frac{m}{2}a_0 \;,\; \sigma = \frac{F'}{S} = \frac{ma_0}{2S} = \frac{\rho LSa_0}{2S} = \frac{\rho La_0}{2}$$

(b)



$$T = \frac{mx}{L}a_0$$

$$\delta = \frac{Tdx}{SY}$$

$$\Delta\ell = \int_0^L \frac{Tdx}{SY} = \int_0^L \frac{mxa_0}{SYL}dx = \frac{ma_0L}{2SY} = \frac{\rho LSa_0L}{SYL} = \frac{\rho L^2a_0}{2Y}$$

6. $F = \eta A\frac{\Delta V}{\Delta Z}$

$$9.8 = 5\times10^{-3}\pi \times 796\times10^{-3} \times 200 \times 10^{-3} \times \frac{v}{2\times10^{-3}}$$

$v = 7.841$ m/s

7. $F = \eta A\frac{\Delta V}{\Delta Z}$ where $F = mg\sin\theta = 130 \times \frac{5}{13}$

$$130 \times \frac{5}{13} = \eta \times1\times1\times \frac{0.5}{0.15\times10^{-2}}$$

$$\eta = \frac{50\times0.15\times10^{-2}}{0.5} = 0.15$$

8. $d\tau = r\eta \left(2\pi r dr \frac{\omega r}{h}\right)$

$\tau = \int_0^r \frac{\eta}{h} 2\pi\omega r^3 dr$

$\tau = \int_0^{d/2} \frac{\eta\omega}{h} 2\pi \ r^3 dr$

$\tau = \left[\frac{2\pi\eta\omega}{h} \ \frac{r^4}{4}\right]_0^{d/2}$

$\tau = \frac{2\pi\eta\omega}{h} \ \frac{d^4}{4 \ x \ 16}$

$\tau = \left(\frac{\pi\eta\omega d^4}{32h}\right)$

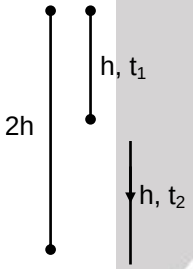
PART - III

1. Stress in wire B $= \frac{mg}{3\pi r_B^2}$

Stress in wire A $= \frac{4mg}{3\pi r_A^2}$

if $\frac{mg}{3\pi r_B^2} = \frac{4mg}{3\pi r_A^2}$ either wire will break.

2.



Time t_1 will be more then t_2 because speed increases from zero to terminal speed in t_1 duration and ball covers a distance h.

Work done against viscous force depends on magnitude of viscous force and displacement ball.

viscous force increases from zero to maximum value and then remains constant

3. $W = -\Delta U = \frac{1}{2}Kx^2 = \frac{1}{2}\frac{AY}{L}\ell^2$

PART - IV

1. $K = \frac{AY}{\ell} = \frac{4\times10^{-4} \ \times \ 2 \ \times \ 10^{10}}{4} = 2 \times 10^6$

$\omega = \sqrt{\frac{K}{m}} = 100$

2. $W(h + x) = 1/2\ kx^2$

$100(0.99 + x) = \frac{1}{2} \times 2 \times 10^6 \times x^2$

$10^4 x^2 - x - 0.99 = 0$

$100\ x\ (100x - 1) + 0.99\ (100\ x - 1) = 0$

$x = \frac{1}{100}\ m = 1\ cm$

3. $x = \frac{PL}{AY}$

$\sigma = \frac{P}{A} = \frac{xY}{L} = \frac{10^{-2} x\ \ 2\ \ x\ \ 10^{10}}{4} = 5 \times 10^7\ N/m^2$

4. $K_1 = 10^6$, $K_2 = 2 \times 10^6$

$K_{eq} = \frac{2 \times 10^6 x 10^6}{3\ \times\ 10^6} = \frac{2}{3}\ \times\ 10^6$

$\omega = \sqrt{\frac{2\ \times\ 10^6}{3\ \times\ 600}} = \frac{100}{3}$

5. Total weight $= 1000 + w$

weight on each rod $= \frac{1000 + w}{4}$

stress $= \frac{1000 + w}{4\ \times\ 4\ \times\ 10^{-4}} = 9 \times 10^6 \quad \Rightarrow \quad w = 14400 - 1000 = 13400\ N$

No. of persons are $= \frac{1340}{50} = 26$

6. $F = \eta A \frac{du}{dy}$

as $u = C_1 y + C_2$

at $y = 0$, $u = 0$ hence $C_2 = 0$

$\frac{du}{dy} = C_1$

$F = \eta A\ C_1$

$2 = 10^{-2} \times 1\ C_1$

$C_1 = 200$

7. $u = c_1 y + c_2$

$c_1 = 200$, $c_2 = 0$

$u = 200 \times 2 \times 10^{-2} = 4\ m/sec$

8. $y = 0$, $u = 0$ $C_3 = 0$

$y = 2\ cm$, $u = 2\ m/sec$

$2 = C_1\ 4 \times 10^{-4} + C_2\ 2 \times 10^{-2}$ ....(1)

$\frac{du}{dy} = 2C_1 y + 2$

$F = \eta\ A\ \frac{du}{dy}$

at $y = 2cm$, $F = 2\ N$

$2 = 10^{-2} \times 1 \times [2 \times 2 \times 10^{-2} C_1 + C_2]$
$4 \times 10^{-4} C_1 + 10^{-2} C_2 = 2$ .....(2)
$4 \times 10^{-4} C_1 + 2 \times 10^{-2} C_2 = 2$ .....(1)
on solving
$C_2 = 0$ & $C_1 = 5000$

9. $\frac{du}{dy} = \frac{F}{\eta A} = \frac{2}{10^{-2} \times 1} = 200$

10. $\frac{du}{dy} = 2C_1 y + C_2$

at $y = 0$, $\frac{du}{dy} = C_2 = 0$

EXERCISE-3

PART - I

1. Terminal velocity $v_T = \frac{2r^2 g}{9\eta} (\rho_s - \rho_L)$

and viscous force $F = 6\pi\eta r v_T$
Viscous force is the only dissipative force. Hence

$\frac{dQ}{dt} = F v_T = (6\pi\eta r v_T)(v_T) = 6\pi\eta r v_T^2$

$= 6\pi\eta r \left\{\frac{2}{9}\frac{r^2 g}{\eta}(\rho_s - \rho_L)\right\}^2 = \frac{8\pi g^2}{27\eta} (\rho_s - \rho_L)^2 r^5 = \frac{dQ}{dt} \propto r^5$

2. $\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{yA/\ell}{m}} = \sqrt{\frac{yA}{\ell m}} \Rightarrow \sqrt{\frac{(n \times 10^9) \times (4.9 \times 10^{-7})}{1 \times 0.1}} = 140 \Rightarrow n = 4.$

3. In equilibrium,
$mg = qE$
In absence of electric field,
$mg = 6\pi\eta r v$
$\Rightarrow$ $qE = 6\pi q r v$
$m = \frac{4}{3}\pi R r^3 d. = \frac{qE}{g}$

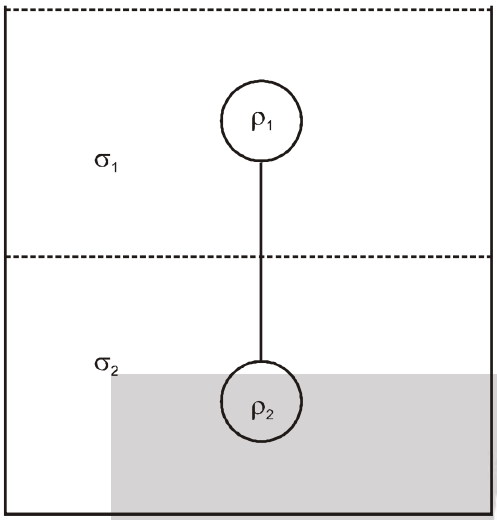
$\frac{4}{3}\pi \left(\frac{qE}{6\pi\eta v}\right)^3 d = \frac{qE}{g}$

After substituting value we get,
$q = 8 \times 10^{-19}$ C **Ans.**

4. $Y = \frac{\left(\frac{F}{A}\right)}{\frac{\Delta\ell_1}{L}}$ ...(i) $Y = \frac{\left(\frac{F}{4A}\right)}{\frac{\Delta\ell_2}{2L}}$ ...(ii)

$\frac{\Delta\ell_1}{\Delta\ell_2} = 2$

6.



For floating

$(\rho_1 + \rho_2)V = (\sigma_1 + \sigma_2)V$

$\rho_1 + \rho_2 = \sigma_1 + \sigma_2$

since strings in taut so

$\rho_1 < \sigma_1 \qquad \rho_2 > \sigma_2$

$$V_P = \frac{2}{9}\frac{r^2(\sigma_2 - \rho_1)g}{\eta_2}$$

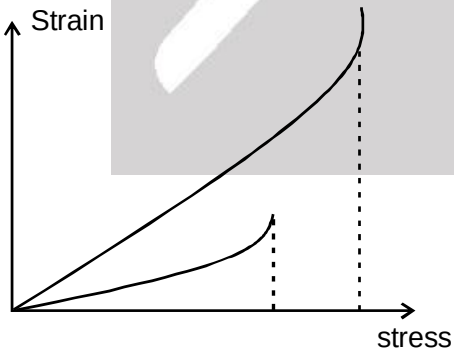
$$V_Q = \frac{2}{9}\frac{(\sigma_1 - \rho_2)g}{\eta_1}$$

since $\sigma_2 - \rho_1 = -(\sigma_1 - \rho_2)$

$$\left|\frac{V_P}{V_Q}\right| = \frac{\eta_1}{\eta_2}$$

$\vec{V}_P \cdot \vec{V}_Q < 0$ because V_P and V_Q are opposite

7.



Breaking stress of P is more than Q so P is more ductile

$$\text{strain} = \frac{1}{Y}\text{stress}$$

$Y_p < Y_Q$

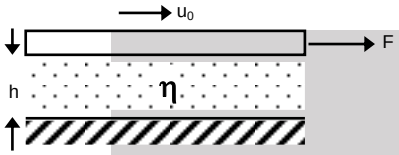
8. $6\pi\eta rv + \rho_L Vg = \rho_0 Vg$

$$\frac{v_P}{v_Q} = \frac{(\rho_P V_P - \rho_L V_p)g}{6\pi\eta_p r_p} \times \frac{6\pi\eta_Q r_Q}{(\rho_Q V_Q - \rho_L V_Q)}$$

$$= \frac{r_p^3(8-0.8)}{\eta_p . r_p (8-1.6)} \times \frac{r_Q . \eta_Q}{r_Q^3}$$

$$= \left(\frac{r_p}{r_Q}\right)^2 \times \left(\frac{\eta_Q}{\eta_p}\right) \times \left(\frac{7.2}{6.4}\right) = 4 \times \frac{7.2}{6.4} \times \frac{2}{3} = 3$$

9.



$$F = \eta A \frac{dv}{dy}$$

$$= \eta A \frac{u_0}{h}$$

10. $v_0 = \frac{J}{m} = \frac{1}{0.4} = \frac{5}{2}$ m/s

$$s = v_0 \int_0^{\tau} e^{-t/\tau}\, dt$$

$$= v_0\tau\,(1 - e^{-1})$$

$$= \left(\frac{5}{2}\right)(4)(0.63) = 6.30 \text{ m}$$

PART - II

1. Retarding force acting on a ball falling into a viscous fluid

$F = 6\pi\eta Rv$

where R = radius of ball,

v = velocity of ball,

and η = coefficient of viscosity

$\therefore$ $F \propto R$ and $F \propto v$

2. Young's modulus = $\frac{\text{Strees}}{\text{Strain}}$

$\Rightarrow$ Strain = $\frac{S}{Y}$

$$\frac{\text{Energy stored in wire}}{\text{Volume}} = \frac{1}{2} \times \text{Stress} \times \text{Strain}$$

$$= \frac{1}{2} S \times \frac{S}{Y} = \frac{S^2}{2Y}$$

3. Terminal speed of spherical body in a viscous liquid is given by

$$v_T = \frac{2r^2(\rho - \sigma)g}{9\eta}$$

where ρ = density of substance of body,

σ = density of liquid.

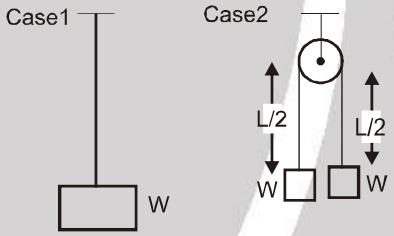
From given data

$$\frac{v_T(Ag)}{v_T(Gold)} = \frac{\rho_{Ag} - \sigma_l}{\rho_{Gold} - \sigma_l}$$

$$\Rightarrow \quad v_T\,(Ag) = \frac{10.5 - 1.5}{19.5 - 1.5} \times 0.2 = \frac{9}{18} \times 0.2$$

$= 0.1$ m/s

4. Let us consider the length of wire as L and cross-sectional area A, the material of wire has Young's modulus as Y.



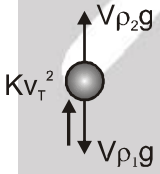
Then for 1st case $Y = \frac{W/A}{\ell/L}$

For 2nd case, $Y = \frac{W/A}{2\ell'/L}$

$\therefore \quad l' = l/2$

So, total elongation of both sides = $2\ell' = \ell$

5. The forces acting on the ball are gravity force, buoyancy force and viscous force. When ball acquires terminal speed, it is in dynamic equilibrium, let terminal speed of ball is v_T . So,



$V\rho_2 g + kv_T^2 = V\rho_1 g \qquad v_T = \sqrt{\frac{V(\rho_1 - \rho_2)g}{k}}$

6. $F = \frac{Y \;\; A \;\; x}{\ell}$

and $F_2 = \frac{Y \;\; (3A \;) \;\; x}{(\ell/3)} = 9\,F$

7. $V\rho g = 6\pi\eta rv + v\rho_\ell g$

$Vg(\rho - \rho_\ell) = 6\pi\eta rv$

$Vg(\rho - \rho_\ell') = 6\pi\eta' rv'$

$V'\eta' = \dfrac{(\rho - \rho_\ell')}{(\rho - \rho_\ell)} \times v\eta$

$V' = \dfrac{(\rho - \rho_\ell')}{(\rho - \rho_\ell)} \times \dfrac{v\eta}{\eta'}$

$= \dfrac{(7.8 - 1.2)}{(7.8 - 1)} \times \dfrac{10 \times 8.5 \times 10^{-4}}{13.2}$

$v' = 6.25 \times 10^{-4}$ cm/s.

8. $\dfrac{P}{\alpha\Delta\theta} = Y$

$P = Y\alpha\Delta\theta = 2 \times 10^{11} \times 1.1 \times 10^{-5} \times 100$

$= 2.2 \times 10^{8}$ Pa

9. $T = 2\pi\sqrt{\dfrac{\ell}{g}}$

$T_M = 2\pi \sqrt{\dfrac{\ell + \Delta\ell}{g}}$ $\qquad \Delta\ell = \dfrac{Mg\ell}{AY}$

$\dfrac{T_M}{T} = \sqrt{\dfrac{\ell + \Delta\ell}{\ell}}$

$\left(\dfrac{T_M}{T}\right)^2 = 1 + \dfrac{\Delta\ell}{\ell}$

$\dfrac{1}{y} = \left(\left(\dfrac{T_M}{T}\right)^2 - 1\right) = 1 + \dfrac{Mg}{AY}$

$\dfrac{1}{y} = \left(\left(\dfrac{T_M}{T}\right)^2 - 1\right)\dfrac{A}{Mg}$

10. $\Delta P = \dfrac{mg}{a}$

$K = -\dfrac{\dfrac{mg}{A}}{\dfrac{4\pi r^2 dr}{\dfrac{4}{3}\pi r^3}}$

$\dfrac{dr}{r} = -\dfrac{mg}{3KA}$

HIGH LEVEL PROBLEMS (HLP)

SUBJECTIVE QUESTIONS

1. $\rho = 7.6\ g/cm^3$

$\ell + \Delta\ell = 90$ cm

$$\Delta\ell = \frac{w\ell}{AY} \quad ....(1)$$

$$\Delta\ell - 0.18 = \frac{\left(w - \frac{w}{\rho_0}\rho_\ell\right)\ell}{AY} \quad ....(2)$$



$$\frac{\Delta\ell - 0.18}{\Delta\ell} = 1 - \frac{\rho_\ell}{\rho_0}$$

$$1 - \frac{0.18}{\Delta\ell} = 1 - \frac{1}{7.6}$$

$\Delta\ell = 7.6 \times 0.18$

then $\ell + \Delta\ell = 90$

$\ell = 90 - 7.6 \times 0.18 = 88.632$ cm

2. r = 2 mm

$$\left(\frac{F}{2A}\right)_{max} = 1.5 \times 10^9$$

$F = 1.5 \times 10^9 \times A \times 2$

$= 1.5 \times 10^9 \times \pi(2 \times 10^{-3})^2 \times 2$

$= 18840 \times 2 = 3.77 \times 10^4$ N

3. $$\frac{4}{3}\pi R^3 = \frac{4}{3}\pi r^3 \times 8$$

$R = 2r$

$V \propto r^2$

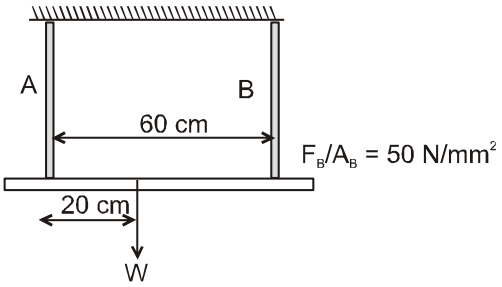
Hence 20 cm/sec.

4. $$v = \frac{2}{9\eta}r^2(\rho_0 - \rho_a)\ g$$

$$0.5 = \frac{2}{9\eta} \times 0.8 \times 100$$

$$\eta = \frac{320}{9} = 35.55 \text{ poise}$$ **Ans.**

5.



$Y_A = 2 \times 10^5\ N/mm^2$

$Y_B = 9 \times 10^4\ N/mm^2$

$F_A \times 20 = F_B \times 40$

$F_A = 2F_B$

$F_A + F_B = 5000\ g$

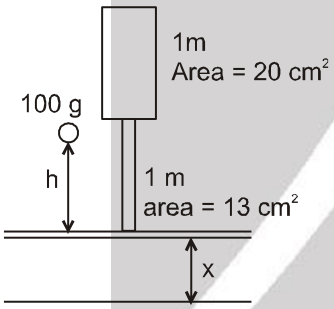
$F_B = \frac{5000\ g}{3}$, $F_A = \frac{10000\ g}{3}$

$\sigma_B = 50 = \frac{F_B}{A_B}$ $\qquad A_B = \frac{50 \times 10^3}{3 \times 50} = \frac{1000}{3}\ mm^2$

$\Delta\ell = \frac{F_B \ell}{A_B Y_B} = \frac{F_A \ell}{A_A Y_A}$, $\qquad \sigma_A = \frac{F_A}{A_A} = \frac{Y_A}{Y_B}\sigma_B = \frac{20}{9} \times 50 = \frac{1000}{9}\ N/mm^2$

$\sigma_A = \frac{1000}{9} = \frac{10^5}{3\ A_A}$ $\qquad A_A = 300\ mm^2$

6.



$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$, $k_1 = \frac{A_1 Y}{\ell}$, $k_2 = \frac{A_2 Y}{\ell}$, $\ell = 1\ m$

$k = \frac{k_1 k_2}{k_1 + k_2} = \frac{A_1 A_2}{(A_1 + A_2)} \frac{Y}{\ell}$

$k = \frac{52}{33} \times 10^8$

$F = 50 \times 10^6 \times 13 \times 10^{-4} = 650 \times 10^2\ N$

$F = kx$

$x = \frac{165}{4} \times 10^{-5}\ m$ $\qquad mg\ (h + x) = \frac{1}{2} kx^2 = \frac{F^2}{2k}$

on solving $\qquad h = 1.33\ cm$

7. $\tau = Fr$

$$F = \frac{\eta(2\pi r_i) \times \omega r_i \times \ell}{r_0 - r_i}$$

$$\Rightarrow \quad \tau = 2\pi\eta r_i^3 \times \frac{\omega \times \ell}{(r_0 - r_i)}$$

on solving and putting values

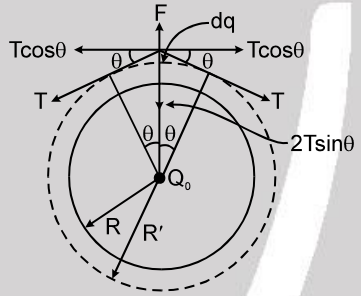
$\eta = 0.77$ N-s/m^2

8. Considering an element of angular width 2θ -

$dq = \lambda R' \cdot 2\theta \quad \Rightarrow \quad F = 2T \sin\theta$

$$\frac{kdq.\ Q_0}{R'^2} = 2T\sin\theta \quad \Rightarrow \quad \frac{k\lambda R' \times 2\theta.\ Q_0}{R'^2} = 2T\sin\theta$$

if θ is small, then $\sin\theta = \theta$



further $\frac{k\lambda Q_0}{R'} = T$

But $Y = \frac{\text{stress}}{\text{strain}}$, strain $= \frac{2\pi(R' - R)}{2\pi R} = \frac{R' - R}{R}$

$$Y = \frac{T}{a(R' - R)} \Rightarrow R' - R = \frac{k\lambda Q_0}{R'AY}$$

$$Y = \frac{T}{A\frac{\Delta R}{R}} \Rightarrow \Delta R = \frac{TR}{AY} = \frac{k\lambda Q_0 R}{R'AY} = (R \simeq R')$$

9. $\frac{mg}{A} = Y\,\alpha\,\Delta\theta \Rightarrow m = \frac{Y\alpha\Delta\theta A}{g} = 4\text{kg}$

$U = \frac{F^2 L}{2AY} = 10^{-3}$ J